



TECHNICAL MEMORANDUM:
**AN OPTIONAL METHOD FOR ADJUSTING
THE RECOMMENDED ILLUMINANCE FOR
VISUALLY DEMANDING TASKS WITHIN IES
ILLUMINANCE CATEGORIES P THROUGH Y
BASED ON LIGHT SOURCE SPECTRUM**
AN AMERICAN NATIONAL STANDARD



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Publication of this document
has been approved by the IES.
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be directed to the IES.

**Prepared for IES
IES Vision Science Committee.**



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1.0 Introduction and Scope

1.1 Introduction

This Technical Memorandum (TM) addresses how the spectral power distribution (SPD) of light sources can be incorporated into the IES Illuminance Determination System for visual tasks that are categorized as P through Y within interior lighting applications (refer to *ANSI/IES LS-8-20, Lighting Science: Vision – Perceptions and Performance*, Section 6.2¹). This TM describes how a light source's SPD affects pupil size, visual acuity, and visual efficiency under interior lighting conditions, and establishes an optional calculation method, Equivalent Visual Efficiency (EVE), for adjusting IES recommended illuminance targets under limited circumstances when visual tasks require the ability to discern achromatic visual detail to ensure speed and/or accuracy in the performance of the visual task. EVE is a method for achieving a balance between light level and SPD that results in *maintaining* equal visual acuity; it is not intended to improve visual acuity or visual performance. The calculation makes no claim as to any effect on overall visual performance, other than to state that the use of this calculation ensures equivalent visual efficiency for visually demanding tasks. The calculation is optional, and its use should be balanced with other lighting design criteria and objectives.

It is acknowledged that this version of TM-24 continues to use an empirically derived, physiologically inaccurate model of the pupillary light reflex based on the ratio of scotopic to photopic power of the light source. In conditions typical of interior lighting, long-course changes in pupil size are principally mediated by intrinsically photosensitive retinal ganglion cells (ipRGCs), not the rod photoreceptors. It is expected that the EVE method will be updated when quantities related to ipRGC photoreception are standardized by the IES.

1.2 Scope

This TM documents the fundamentals of visual efficiency, defines the EVE calculation method, and addresses practical concerns for implementation. The EVE method described herein should not be applied if any of these conditions apply:

- The task or tasks are covered by IES Illuminance Categories A through O for the Recommended Illuminance Target¹
- The task is nonvisual in nature
- The task is not visually demanding (see **Section 2.5**)
- The application is exterior
- It is known that there are a high number of occupants with abnormal vision or poor ocular health

Further discussion of the preconditions for application of the EVE method is included in **Section 4.5**.

The method described in this TM is based on how the SPD of a light source affects the ability to discern achromatic visual detail. The SPD can also affect brightness perception, color rendering, discomfort glare, circadian rhythms, and other physiological outcomes; these factors are discussed only in limited detail in this TM. The decision to apply the EVE method is at the discretion of the specifier.

2.0 Overview: Concepts and Definitions

2.1 The Role of Light Source Spectral Power Distribution

Quantifying the effects of lighting on human vision is one of the fundamental requirements of illuminating engineering. This task is accomplished by converting light as an electromagnetic quantity defined in radiometric values to units that have some relationship to human vision. This conversion is accomplished by using the spectral power distribution (SPD) of a light source as a radiometric quantity and weighting that SPD against the spectral response of the human visual system.

2.2 The Spectral Luminous Efficiency Function for Photopic Vision, $V(\lambda)$

The spectral luminous efficiency function for photopic vision, $V(\lambda)$, is used to weight radiant energy according to the sensitivity of the visual system at relatively high light levels. (For more information see *ANSI/IES LS-2-20*,

*Lighting Science: Concepts and Language of Lighting.*²⁾ Many important lighting metrics (e.g., illuminance, luminance, luminous intensity) are based on the $V(\lambda)$ function and are sometimes referred to as photopic quantities.

The $V(\lambda)$ function, standardized by the International Commission on Illumination (CIE) in 1924, was derived from psychophysical experiments where the visual field was restricted to the central 2 degrees, corresponding to the fovea and isolating the response of the cone photoreceptors.³ Foveal vision is responsible for detection of fine detail, and while important, the spectral sensitivity of the fovea can vary from that of full-field vision because the fovea excludes the possible contributions of photoreceptors whose concentrations are primarily peripheral to the central 2 degrees. These include a portion of the S cones, the rods, and the intrinsically photoreceptive retinal ganglion cells (ipRGCs; refer to ANSI/IES LS-7-20, *Lighting Science: Vision – Eye and Brain* for more information⁴⁾). The metrics derived from $V(\lambda)$ are central to illuminating engineering but are often extended beyond the conditions under which they were developed,⁵ and they do not address all aspects of visibility and visual performance, such as the regulation of pupil size.

2.3 Pupil Size, Retinal Image Quality, and Visual Acuity

The importance of pupil size is that it can affect visual acuity, depth of field, and accommodative effort. Of particular concern to lighting practice is the effect of pupil size on retinal image quality and, consequently, visual acuity. (Refer to ANSI/IES LS-8-20 for more information.¹⁾)

At typical interior illuminance levels, relatively smaller pupils act to remove the most non-central aberrant light rays, which for most eyes are the principal cause of blurred images. Consequently, pupil size is one of the principal physiological agents affecting optical quality for most people, and it is the quality of the optical image reaching the retina, rather than the innate retinal resolution that can be a highly important factor affecting the overall resolution of the eye. Thus, confining light to a smaller aperture through smaller pupils increases sharpness of focus as well as depth of

field. These findings apply to the general population with natural healthy eyes and habitual correction.

Optimum visual acuity is typically achieved with pupil size between 2 and 3 mm, although in some circumstances this range may extend slightly higher.⁶ However, those conditions are of sufficiently low luminance levels and pertain to tasks of low spatial frequency, and thus are outside the domain of indicated TM-24 applications. Visual resolution generally decreases as pupil size increases beyond 3 mm. Under the conditions of a full field of view, average pupil diameter changes in an approximate log-linear manner between its maximum diameter of 6 to 7 mm below 1 cd/m² and its minimum size of 2 to 3 mm at 1,000 cd/m² and greater. Thus, changes in pupil size that occur under normal interior illuminance levels and under a full field of view are likely to affect retinal image quality for normal-sighted people.

Visual acuity (VA) refers to the minimum size that can be resolved under specific viewing conditions, such as luminance, contrast, and distance. Visual acuity is a variable unique to each person and varies with age.⁷ The average VA for the general population is slightly better than 20/20 prior to age 50 and worsens with age from that point. An individual's VA is evaluated under prescribed conditions of task luminance and contrast. However, luminance is defined by $V(\lambda)$ and therefore does not account for pupil size variations that occur due to changes in SPD (at fixed light levels) that in turn can affect VA. This is discussed further in **Section 3.2**, and additional review of work in this field is available from Annex B of TM-24-13.⁸ Variations in visual acuity can affect visual efficiency for visually demanding tasks.

More broadly, improvements in visual image quality generally attributed to increased illumination levels could be largely due to the improved retinal image quality resulting from smaller pupils, even though smaller pupils lead to lowered retinal illuminance.⁹⁻¹¹ Within the realm of interior light levels, increasing pupil size under a constant task luminance has been shown to result in reduced visual acuity, even though the retinal illuminance increases.^{12,13} Further, pupil size can be reduced with higher levels of surround illuminance,¹⁴ and may be influenced by psychological factors. Finally,

it is noteworthy to call out exceptions to the pupil size-visual acuity relationship that occur under conditions of high luminance. When task luminance values increase to values of 500 cd/m² or greater, visual acuity can be negatively affected through the introduction of disability glare.^{11,12}

2.4 Visual Efficiency and Visual Effort

Visual efficiency has been defined as "The ability to perform visual tasks easily and comfortably."¹⁵ The onset of visual efficiency (i.e., the point at which tasks become easy and comfortable to perform) occurs when the combination of task size, contrast, and illuminance level, as well as light source SPD, has reached a point such that visual performance is essentially constant and approximately independent of further incremental increases in these variables. That is, visual performance reaches a plateau when the task size reaches a value of about three times the acuity threshold, with other factors remaining constant.¹⁶⁻¹⁹ Threshold conditions, such as VA testing, form the baselines that establish the onset of regions of visual efficiency under varying combinations of lighting and task parameters. (Refer to ANSI/IES LS-8-20 for more information.¹)

Visual effort relates to the relative level of difficulty of performing tasks, with more visual effort expended as the lighting and/or task conditions get closer to the threshold limit. Visual efficiency is therefore achieved when visual effort is minimized and is achieved more rapidly when VA is improved. These concepts are further explored in **Section 4.6** of IES TM-24-13.⁸

2.5 Visually Demanding Tasks

Visual demand is a broad term used to describe the effort exerted by the visual system in performing visual tasks and has, as one of its parameters, the expectations of a performance output. For the purpose of this TM, *visually demanding tasks* are separately defined as *tasks that are based on the ability to discern achromatic visual detail to ensure speed and/or accuracy*. This definition: 1) emphasizes that tasks are based on *visual ability*, thereby acknowledging the independent variable of an individual's visual system; 2) restricts the tasks to those that require *detailed* vision, which when combined with the human visual capability is best described by VA; and 3) further requires that this definition be limited to those

tasks for which the speed and/or accuracy performance might be negatively affected by decrements in the visual ease of performing the task. A commonplace example of a visually demanding task is reading printed material in a work environment, which is dependent on a number of variables, including an individual's VA, the size and contrast of the printed material, and the importance placed on the speed and/or accuracy of performing the task. Reading ultimately relies on the visual capability to discern detail, as generally quantified by visual acuity, and is therefore considered a *visually demanding task*.

The term *visually demanding task* refers to the type of task being performed and is different from the term *visually difficult task*, which describes the interrelation between the task and the person performing the task. For example, if a reading task is sized at 20/40 at a fixed normal reading distance from the eye, then a person with 20/20 vision is likely to be able to perform the task relatively quickly, more accurately, and with less visual effort compared a person with 20/40 vision. In such a comparison, the visually demanding task of reading is held constant; but the visual difficulty of performing the visually demanding task is based on each person's visual capability as illustrated through their differing visual acuities. In this example, the person with 20/20 vision would have reached the onset of visual efficiency, but the person with 20/40 vision would be at their acuity limit, expending more visual effort to perform the task, which in turn would result in slower reading speed and more errors.

Lighting plays a significant role in facilitating performance of visually demanding tasks. The IES Illuminance Determination System provides categories for recommended task illuminance based largely on the task parameters with the goal of ensuring visual efficiency for performance of these tasks by the general population—people with habitual correction of approximately 20/20 and good ocular health.

In the Illuminance Determination System, Categories A through O comprise tasks that do not require detailed vision, and generally do not benefit from improved VA. The demarcation point for when the ability to discern detail becomes relevant is in illuminance categories P

through Y, where the general description for Category P is “Common, relatively small-scale, more cognitive or fast-performance visual tasks,” characterized as tasks for which “visual performance is typically daily life- and work-related, including much reading and writing of hardcopies and electronic media consecutively and/or simultaneously.”²⁰ Thus, these categories include visually demanding tasks. However, some visual tasks within these categories have other criteria and are therefore not visually demanding tasks. Examples of IES Category P through Y visual tasks that are not visually demanding tasks include:

- Nonvisual tasks, such as lighting for plant growth, camera and film lighting, and process lighting (such as curing)
- Tasks specific to color appraisal or identification
- Lighting for attention, such as focal lighting, retail show window lighting, building façade lighting, and theatrical lighting
- Self-illuminated tasks, such as typical computer screens (those with positive polarity)
- High-activity areas, as found in exercise rooms and fitness facilities

Visually demanding tasks are a subset of those tasks that fall within IES illuminance categories P through Y, those where the task requires the ability to discern achromatic detail to ensure speed and/or accuracy. Tasks that fall into these categories occur in many buildings, such as offices, health care facilities, educational facilities, libraries, factories, and warehouses. It is important to note that sometimes, the performance (speed and accuracy) of visually demanding tasks is not determined by visual performance, but by cognitive ability.²¹ Visual performance should not be confused with task performance. (Refer to ANSI/IES LS-8-20 for more information.¹)

2.6 Visual Performance of Visually Demanding Tasks

Visual performance of visually demanding tasks is generally measured by the speed and/or accuracy of performing the task. For a task, performance is primarily dependent on the combination of three independent variables: 1) illuminance level and SPD; 2) task size and contrast; and 3) a person’s specific visual capability,

generally determined by their VA. When visual tasks are close to a person’s VA threshold limit, the capability to perform these tasks can be affected by changes in their ability to discern detail resulting from changes in lighting or task parameters, and as a consequence, the zone of visual efficiency may be affected.

Improvements in VA result in improved visual efficiency because tasks that were originally at the margin of one’s ability to discern detail become easier; that is, the previously marginal task can now be performed with less visual effort. However, this does not necessarily mean that improvements in VA will lead to increased visual performance of tasks whose detail are of a much larger size or higher contrast than the threshold limit. A simple example of this occurs when a person’s VA improves from 20/30 to 20/20 (through better lighting or new spectacles)—the ability to read a small footnote in 8-point font is made easier, and visual performance might improve for the footnote, but the speed and/or accuracy of reading a normal 10-point font size might not be affected. Conversely, as visual tasks become more difficult and reach a point of being below the onset of visual efficiency, visual effort increases, and the speed and/or accuracy of performing the task might suffer. Thus, if one lighting environment provides a higher degree of visual efficiency—and therefore allows tasks to be performed with less visual effort—then that lighting environment can be considered visually better due to the relative ease of performing some tasks, even though the improvement may not necessarily manifest itself through measurable improvements in visual performance.

For interior lighting applications, the visual acuity of the various individuals working in a given space is unknown, and these same people are performing a variety of visual tasks. If two lighting SPDs are compared at different illuminance levels, and the resulting VAs are equivalent under each combination—with all other conditions (e.g., light distribution, perceived glare, color quality, room reflectances, and tasks) being equal—then the ability to discern detail will also be maintained, as will the zone of visual efficiency. Thus, if visual acuity is maintained, visual efficiency and visual performance of visually demanding tasks will also be maintained.

The connection between visual acuity, visual efficiency, and overall visual performance is illustrated in **Figure 2-1**. The diagram shows the experimentally determined visual performance for a group of subjects for a reading task in words per second as it relates to the size Visibility Level (VL),¹⁸ which is the ratio of the letter size to the threshold size. Visual performance begins where the ability to correctly identify the detail of a task becomes statistically significant and is referred to as the threshold, or acuity, point. From this point, visual performance generally rises rapidly through the escarpment range, and then begins to change its form through the performance knee region until it reaches the plateau range where the independent variable (task size, task contrast, illuminance level, or SPD) has little effect on performance. The point where the plateau begins is also the point at which visually efficiency begins (where tasks become easily and comfortably performed). The general shape of visual performance curves as a function of contrast or illuminance level are similar, although in the case of tasks of very small size and very low contrast, a flat plateau might never be reached as illuminance levels are increased.

2.7 The Aging Visual System

Pupil size range and steady-state pupil size are generally smaller for the aging eye at all light levels; these effects are known as *senile myosis*. The aging eye also typically contains relatively more aberrations than the

younger eye; therefore, it is possible that in terms of effects on visual performance, the relatively smaller change in pupil size for the older person from a given light source could result in quite similar performance effects compared to those of a younger age group. Further, the eye's lens develops a relative increase in absorption for the bluer part of the visible spectrum. Compared to the 25-year-old, there is about a 60% decrease in transmittance at 480 nm,^{22,23} resulting in a relatively larger impact on melanopic or scotopic quantities as compared to photopic quantities. CIE S026/E:2018 defines a standardized system for adjusting photoreceptor responses for age, allowing SPD-specific age-adjusted responses to be determined.²⁴

3.0 Basis of the Equivalent Visual Efficiency (EVE) Method

3.1 Overview

The historical record of research on SPD and its relationship to visual acuity and visual efficiency includes many studies spanning many decades. Some specific work that is foundational to the EVE method is discussed in this section. Additional background information may be found in Annex B of IES TM-24-13.⁸

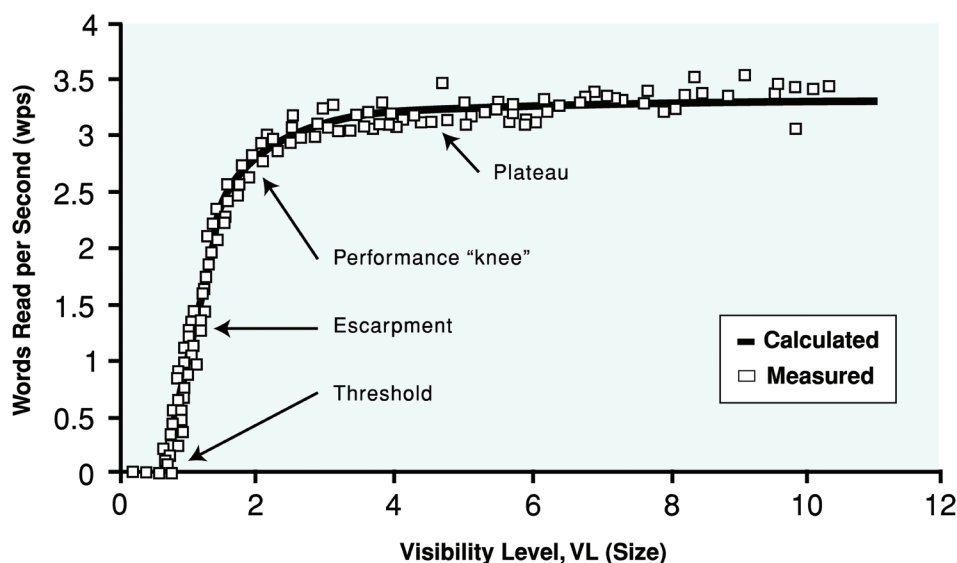


Figure 2-1. A typical visual performance curve showing the regions of visual performance behavior.¹⁸

3.2 Spectral Sensitivity of the Pupillary Light Reflex

The spectral sensitivity of the pupillary light reflex, which controls pupil size, is not the same as $V(\lambda)$. This was first reported in the 1960s,²⁵⁻²⁷ with work continuing through the 1990s finding greater sensitivity to short-wavelength visible radiation with a full unobstructed field of view.^{9,10,28,29,30} (When restricted to a 2-degree field of view, matching the genesis of $V(\lambda)$, pupil size variation can be accounted for with $V(\lambda)$ -based metrics, but this is not typical in an architectural environment.) At the time, pupil response was attributed to the rod photoreceptors and the scotopic visual efficiency function, $V'(\lambda)$, with the empirically derived scotopic-to-photopic ratio (expressed as S/P) used to explain related effects. However, after discovery of intrinsically photosensitive retinal ganglion cells (ipRGCs), subsequent research has confirmed that these photoreceptors are instrumental in mediating pupil size, in combination with other photoreceptors that provide extrinsic signaling.³¹⁻⁴² All three types of photoreceptors work in concert to control pupil size, responding differently to different intensities and durations of light exposure. These signals project to the brain's olivary pretectal nucleus (OPN), not the visual cortex. A complete review of the physiology of the pupillary light reflex is outside the scope of this document.

To date, a complete model of phototransduction for the pupillary light reflex, accounting for all photoreceptors in the eye and their relative contributions in different visual conditions, does not exist. However, it is clear that short-wavelength radiant energy, outside the peak of the $V(\lambda)$ function, is critical for regulating pupil size. (Note that pupil size may also be influenced by contrast and light distribution²¹; for example, discomfort glare is correlated with reduced pupil size).⁴³

The discrepancy between $V(\lambda)$ and the spectral sensitivity of the pupillary light reflex means there are two lighting-based pathways that manipulate pupil size (and potential downstream effects, such as visual acuity): changing the total radiant flux (for example, by increasing the illuminance level) and changing the SPD (see **Figure 3-1**). Both pathways result in changes to the radiant energy delivered to the ipRGCs and other photoreceptors responsible for the pupillary light reflex. Consequently, there is a possible tradeoff between

photopic light levels and the spectral properties of light sources to maintain pupil size. For example, a light source with relatively greater short-wavelength content, resulting in smaller pupils and improved visual resolution of detail at the same photopic light level, allows the lighting design to utilize reduced photopic light levels to maintain equal pupil size and visual efficiency. Conversely, if a light source had relatively less short-wavelength content, the result would be larger pupils and possibly reduced visual acuity and/or visual efficiency under the conditions of equal light levels. In certain situations, depending on visual difficulty of the task and position relative to the plateau of visual efficiency, this could result in reduced visual performance and/or more visual effort required to perform the task.

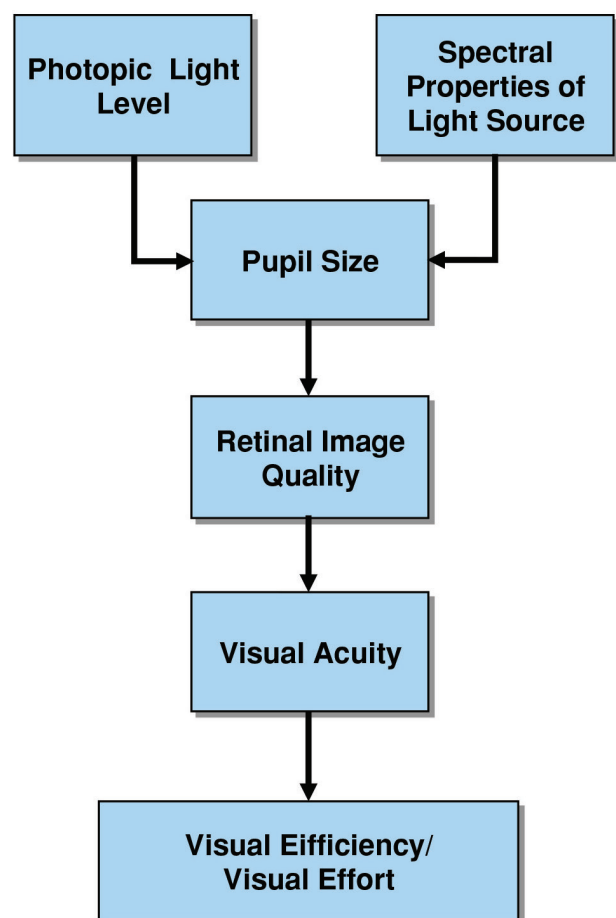


Figure 3-1. Conceptual diagram showing how both light level and SPD contribute to visual efficiency when performing visually demanding tasks in typical interior lighting applications where the eye is exposed to a full field of view.

Absent a complete physiologically accurate model of phototransduction for the pupillary light reflex, the EVE method relies on an empirical relationship correlating pupil size variation with both $V(\lambda)$ and SPD, via the S/P value. This relationship, determined in 1992 before the discovery of the ipRGCs, does not predict absolute pupil size, but rather shows how SPD and light level—as derived from $V(\lambda)$ —combine to predict the light-driven average variation in pupil size. The specific dependency was shown empirically to correlate with the quantity $P(S/P)^x$, where: P is the photopic light level; S/P is the scotopic-to-photopic ratio expressed as a fraction; and x is an exponent that depends on the specific conditions of observation. For the viewing condition of a full field of view in a spectrally flat white and object-free environment, and with subjects viewing a fixation point at 1 meter away, the exponent was empirically measured to have the value 0.78 ± 0.03 . This result leads to the conclusion that two different SPDs of light levels P_1 and P_2 would produce equal pupil size when the condition $P_1(S/P)_1^{0.78} = P_2(S/P)_2^{0.78}$, where: subscripts 1 and 2 represent light sources that differ in their spectral characteristics; P represents the conventional lighting metric based on $V(\lambda)$ (e.g., luminance, illuminance); and S/P is the S/P value of the two SPDs being compared.

Empirically deriving formulas from available data is an important scientific technique, but they are subject to concerns about external validity. In this case, the model was derived based on a limited number of SPDs, which do not represent the range of possible SPDs that could be used in architectural lighting today or in the future. It has been effective for characterizing pupil size in other experiments,^{10,29,30,44,45} although because it is not based on the correct physiology, there is potential for substantial error, especially with future narrowband SPDs (e.g., laser diodes). For example, two SPDs with the same ipRGC stimulation and illuminance (i.e., M/P value) can differ in S/P by as much as 0.70, which is a substantial difference in EVE method outcome. This situation can be remedied with an improved model of photosensitivity for the pupillary light reflex. The ratio of melanopic to photopic radiation is a potential model that has been considered⁴⁶; it is more accurate physiologically, but standardized response functions for ipRGCs are

just being finalized. The peak photosensitivity of the ipRGCs (490 nm)^{24,47} is similar to that of rod photoreceptors (507 nm), which is one reason why S/P can be a relatively useful proxy for understanding pupillary light reflex for broadband illuminants, even if it is physiologically incorrect.

Experimental results show that the use of light sources with relatively higher short-wavelength content in their SPD (which results in smaller pupil size) results in a correlating improvement in VA. Berman et al.³⁰ examined acuity and contrast sensitivity with pupil size regulated by varying the spectral content of the ambient lighting, under the conditions of a full field of view and natural binocular vision. They concluded that for light levels typical of interior environments, the improved resolution resulting from smaller pupils plays a more important role than the increased retinal illuminance that occurs with a larger pupil. By extension, this suggests that the spectral benefit could continue until the pupil reaches its minimum size.

3.3 Adjusting Light Source SPD to Improve Visual Performance

The EVE method is intended to work as approximation for maintaining the ability to discern achromatic visual detail under different light source SPDs at different illuminance levels, based on the premise that equal pupil size is the principal controlling physiological condition for maintaining visual acuity. The EVE method's quantification for the illuminance-level/SPD tradeoff for visually demanding tasks is determined by the pupil size spectral factor $P(S/P)^{0.78}$, with the restrictions that this applies to relatively uniform interior lighting conditions where the eyes are exposed to a full field of view with complete adaptation, for the general population with good ocular health.

Figure 3-2 illustrates the general shape of the visual performance curve in **Figure 2-1** (see **Section 2.6**) as a function of illuminance level for two hypothetical light sources with different SPDs.

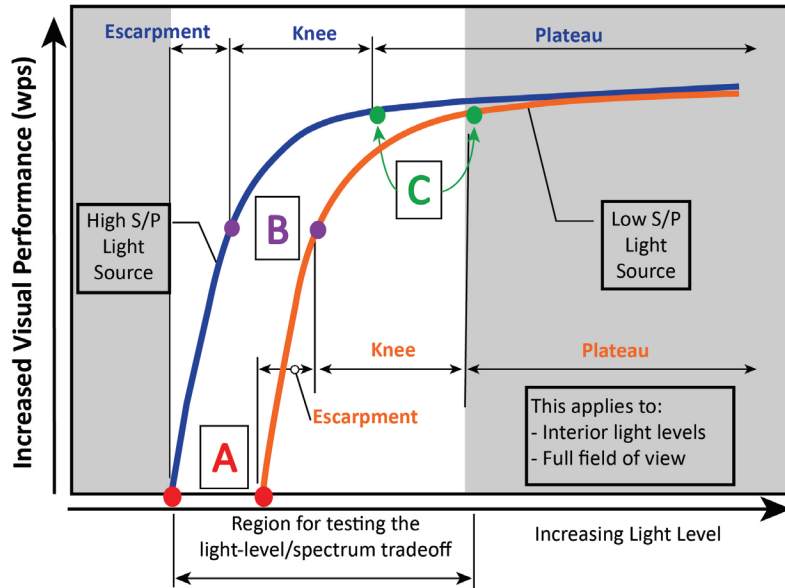


Figure 3-2. Two typical visual performance curves are shown for a hypothetical visually demanding task under two illuminants of differing S/P values. These curves are based on the equivalent ability to discern achromatic visual detail resulting from spectrally driven pupil size. The visual performance curves for the light sources are approximately identical in shape. The two curves demonstrate the illuminance-level/SPD tradeoff: throughout the region where illuminance levels affect the ability to discern visual detail under the lower-S/P light source, equivalent visual performance can be achieved under a higher-S/P light source at a lower illuminance when compared to the lower-S/P light source (points A, B, and C).

The rationale for an illuminance-level/SPD tradeoff is:

1. If spectrally driven pupil size results in equal-sized pupils under a light source with a relatively greater S/P value, but at a lower illuminance level compared to a source with a lower S/P value, then the ability to discern achromatic visual detail would also be equal under the compared light sources.
2. Given constant task parameters of size and contrast and with the sole independent variable of illuminance level, the performance curves under the two compared light sources would be identically shaped throughout the performance curve due to the equal ability to discern achromatic visual detail throughout the range of values for which the target is visibly discerned.
3. In the region where both performance curves have plateaued, there would be negligible differences in performance between the two light sources from SPD or illuminance.

For a given visually demanding task with fixed size and contrast, the EVE method says that equivalent visual performance occurs at a reduced illuminance level for the source with the relatively higher S/P value (for example, see points A, B, and C in **Figure 3-2**), up to the point where the two curves meet in the plateau region. Beyond the point where the two curves merge, neither illuminance level nor SPD has a significant effect on visual performance, although a tradeoff between the two is still possible to achieve the same visual efficiency. The illuminance level where the plateau begins depends on the task characteristics.

Several experiments have been conducted to explicitly examine the illuminance-level/SPD tradeoff at the core of the EVE method,^{10,12,13,21,29,44,45,48,49} and some others have directly or indirectly examined the relationship between illuminance level, SPD, and visual performance.⁵⁰⁻⁵⁵ To evaluate the illuminance-level/SPD tradeoff, an acuity-based task with sufficient sensitivity is presented in

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conditions where a reduction in illuminance leads to a reduction in visual performance, as indicated in **Figure 3-2**. A fixed position for the viewer ensures that viewers cannot compensate for changes in acuity by moving their head, which changes the task size, and thus difficulty, which in turn can obscure the illuminance level/SPD tradeoff. The likelihood of a confirmatory finding increases with a full field of view and complete adaptation, reflecting current understanding of the pupillary light reflex mechanism—although greater understanding of how a reduced field of view and incomplete adaptation influence the illuminance-level/SPD tradeoff could be informative. Finally, visual performance is only one part of task performance; thus, experiments in this area need be careful to control for variables such as comprehension or motor skills, which alter task performance but not visual performance. Important factors for evaluating this body of literature are discussed further in Annex A of TM-24-13.⁸

Several studies have shown that the pupil size spectral factor $P(S/P)^{0.78}$ correlates well with visual acuity and/or visual performance of visually demanding tasks, but others have reached different conclusions, leading to some confusion about whether and how light sources SPD might play a role in lighting practice. The differences between various research findings can be explained by the experimental designs, research protocols, and measurements of performance that have been used. The collective results help validate the EVE method while also indicating where the method should and should not be expected to be effective—and therefore helping to establish parameters for its recommended use.

The studies that have sought to explicitly test the EVE method have generally confirmed the $P(S/P)^{0.78}$ relationship. Two studies found the $P(S/P)^{0.78}$ factor to be conservative for high contrast tasks. Navvab tested the word reading acuity of approximately 100 college-age subjects with both horizontal and tilted task orientations.¹² Two fluorescent lamps, with S/P values 1.3 and 2.2, were used, and while the different photopic illuminances at the subjects' eyes were approximated using the $P(S/P)^{0.78}$ formula, the measured performance was significantly better under the higher- S/P light source even though the photopic illuminance had been reduced. Berman and colleagues tested standard letter

near visual acuity for 27 elementary school students comparing fluorescent light sources with S/P values of 1.3 and 2.2, respectively.⁴⁶ Equality of average near visual acuity occurred when the illuminance level for the higher- S/P condition was reduced by 50%, whereas the $P(S/P)^{0.78}$ formula would predict a reduction of photopic illuminance of only 34% for equivalence.

One study found the $P(S/P)^{0.78}$ formula to over-predict the SPD effect for a low contrast task. Liebel and colleagues tested the applicability of the $P(S/P)^{0.78}$ formula for near-vision reading performance using paper-based, contrast varying tasks under ambient lighting for conditions where the illuminance-level/task combinations were in the performance “knee” region (see **Figure 3-2**).²¹ The study included 47 subjects of ages varying from 18 to 50. The $P(S/P)^{0.78}$ formula predicted reading performance equivalence between a task with luminance of 65 cd/m² illuminated by a fluorescent lamp with S/P of 1.3, and the same task with luminance of 49.5 cd/m² illuminated by a fluorescent lamp with S/P of 2.0. These luminances are correlated to incident illuminances of 220 and 167 lux. The spectral effect was found to exist in the same region of task difficulty for which illuminance also made a difference. Furthermore, as the task was made easier through higher contrast levels, the effect of both illuminance and SPD on performance diminished in parallel. At the lowest contrast levels, this study found that the $P(S/P)^{0.78}$ formula did not fully predict the performance equivalence when the incident illuminances were reduced to 167 lux for the lamp with S/P of 1.3 and to 128 lux for the lamp with S/P of 2.0; in this case, there was still a spectral effect but not as strong as the $P(S/P)^{0.78}$ formula predicts. It is unclear whether any of the error is due to using the S/P proxy for the pupillary light reflex that is dominated by the ipRGC, not the rod photoreceptors.

While the indications from the high contrast studies indicate that the $P(S/P)^{0.78}$ formula is conservative, the realistic condition in lighting applications for which IES Illuminance Categories P through Y are recommended is that tasks are quite varied and could include low contrast tasks. For visually demanding tasks in interior lighting applications, the EVE method, utilizing the $P(S/P)^{0.78}$ formula, should only be used when the baseline P_1 luminance level is 50 cd/m² or greater. (Note that for

some tasks, this background luminance level may fall on the visual performance plateau.)

Boyce et al.⁴⁴ confirmed that when the illuminance level and task combination are such that visual performance has plateaued, as evidenced by no change in visual performance with a reduction in illuminance (and no change in SPD), an illuminance-level/SPD tradeoff cannot be detected for either young or old participants. This may be the case for common office applications, which was the focus of the study, because illuminance levels are typically on the visual performance plateau. Five studies tested illuminance level differences between test conditions that far exceeded the prediction of the $P(S/P)^{0.78}$ formula for tested lamps to compensate for the light level reduction through the differences in their SPDs.^{50,52,53,54,55} Findings in these studies suggest that illuminance level, but not SPD, affected visual performance, which is consistent with the $P(S/P)^{0.78}$ formula.

The effects of aging on vision (see **Section 2.7**) dictate that the effective S/P value for a given source will be less for a 65-year-old than for a 25-year-old—approximately 14% for broadband SPDs, but with considerable variation among all SPDs. A small number of studies have examined how age effects the relationship between spectrally controlled pupil size and visual performance. Berman and colleagues²⁹ found that the performance changes with varying contrast for a Landolt orientation task were about the same for participants aged 61 to 66 and a 20-year-old control group, even though pupil size changes were considerably smaller for the elderly. In contrast, a study by Haymes and Lee⁵⁰ found that, for the normal control subjects with a mean age of 78 years, a statistically significant improvement on maximum contrast sensitivity was measured comparing lighting conditions with S/P values of 1.6 and 1.1; however, this same study did not find a significant effect on visual acuity. In applications where it is known that there are a proportionally high number of occupants with abnormal vision or poor ocular health, care should be taken to ensure that the principles of the visual effects of spectrally driven pupil size apply. Consideration becomes increasingly important when the age of the occupants is older than is typical.

4.0 Equivalent Visual Efficiency (EVE) Method

4.1 Overview

The EVE calculation is a means for incorporating light source SPD into illuminating engineering for interior environments, if certain conditions are met (see **Section 4.5**). The EVE method can be conceptualized as follows: Given a photopic light level A under an illuminant “x” of a known S/P value ($A_{(x)}$), there may be an alternative illuminant “y” with a different S/P value that can provide the same level of visual acuity but with a different photopic light level B ($B_{(y)}$), such that $A_{(x)}$ and $B_{(y)}$ are providing equivalent visual efficiency.

4.2 EVE Calculations

The EVE equations for obtaining equivalent visual efficiency between two different light sources are given below.

EVE, general form:

$$P_2[(S/P)_2^{0.8}] = P_1[(S/P)_1^{0.8}] \quad (4-1)$$

Note: An exponent of 0.8 is used here in lieu of the 0.78 exponent previously proposed. This is a reflection of precision that is expected in this circumstance and is slightly more conservative.

EVE, ratio form, EVE factor:

$$\frac{P_2}{P_1} = \left[\frac{(S/P)_1}{(S/P)_2} \right]^{0.8} \quad (4-2)$$

Note: The above form calculates the resultant photopic metric ratio, or EVE Multiplier, when comparing two light sources of differing S/P values. Values of the photopic metric are not required to calculate this ratio.

EVE, Resultant Photopic Light-Level Form:

$$P_2 = P_1 \left[\frac{(S/P)_1}{(S/P)_2} \right]^{0.8}, \quad (4-3)$$

where:

- Subscripts 1 and 2 represent the two light sources being compared. Subscript 1 refers to the baseline condition, and subscript 2 is the proposed condition.

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- P represents the conventional lighting metric based on $V(\lambda)$ (e.g., luminous flux [lumens], illuminance [lux], or luminance [cd/m^2]).
- S/P is the relevant value of each light source being compared.

Note: This form calculates a new photopic light level, P_2 , given an initial photopic light level P_1 and S/P values of two light sources.

4.3 Example Calculation

A comparison is made between light sources for a visually demanding task in an industrial facility on the basis of visual efficiency. The comparison is between an example LED with $S/P = 1.4$ as the baseline light level source and 1) a typical HPS lamp with $S/P = 0.6$; and 2) LEDs with $S/P = 2.0$. The S/P values for these lamps are examples only. The two comparisons:

- The use of HPS with the lower S/P value would result in larger pupil size and require increased light level relative to the baseline in order to maintain visual acuity and visual efficiency. The amount of additional light, relative to the baseline case of the LED light source, is $(1.4/0.6)^{0.8} = 1.97$, or approximately twice as much.
- The LED2 light source with a higher M/P would result in relatively smaller pupil size, and thereby could be used at a reduced light level relative to the baseline while maintaining the baseline visual acuity/visual efficiency. The calculation is: $(1.4/2.0)^{0.8} = 0.75$, meaning that the resultant target light level is 75% of the baseline light level.

By maintaining the baseline visual efficiency, task performance remains constant regardless of the approach.

4.4 Using the EVE Method with the IES Illuminance Determination System

The EVE method can be applied to the IES Illuminance Determination System (refer to *ANSI/IES LS-8-20, Lighting Science: Vision – Perceptions and Performance*¹). S/P -based EVE multipliers for IES Recommended Illuminance Targets (Table A-4 of *ANSI/IES RP-10-20, Recommended Practice: Lighting Common Applications*²) are determined with a reference S/P value of 1.4—approximately that

of standard incandescent light sources. That is, the recommendations can be adjusted when a light source has an S/P value different from 1.4 by using **Equation 4-2**, as long as all necessary preconditions are met. **Table 4-1** presents EVE-modified Illuminance Targets for Categories P through Y, respectively, which can be applied subject to the constraints described in **Section 4.5**. When implementing the EVE method, the S/P value for the specific light source being considered should be used to determine an appropriate modified illuminance target.

The application of the EVE calculation to the Illuminance Determination System has three general results:

- When lighting specifications utilize light sources with S/P values of 1.3 through 1.5, there will be small changes in visual efficiency that are within IES acceptable tolerance in lighting calculations.
- If light sources with S/P values less than 1.3 are used in interior lighting applications that include visually demanding tasks, there could be a negative effect on the occupants' visual efficiency if the illuminance levels are not increased. The results of not increasing the illuminance levels could manifest itself in decreased visual performance. However, increasing light levels could have a different negative affect, which is added glare. The designer is therefore cautioned to carefully consider all of the design criteria of the application prior to using low- S/P light sources (that is, 1.3 or less) for interior lighting applications with visually demanding tasks.
- If light sources with S/P values greater than 1.5 are used in interior lighting applications that include visually demanding tasks, then illuminance levels can be reduced by the EVE Factor to maintain visual efficiency relative to the baseline. How this may influence the lit environment and building occupants in other ways is discussed in **Section 5**.

It should be noted that correlated color temperature (CCT) and S/P (or M/P) should not be used interchangeably. For example, a commercial light source with a CCT of 3000 K might have an S/P value from as low as 1 to as high as approximately 2.25.⁵⁶ This difference would have large consequences when implementing the EVE method to adjust illuminance targets. With theoretical SPDs,

Table 4-1. EVE-Modified Target Illuminances (Lux) for IES Categories P through Y

Light Source S/P Value	EVE Multiplier	Illuminance Category									
		P	Q	R	S	T	U	V	W	X	Y
0.60	1.97	590	790	980	1,480	1,970	2,950	3,940	5,910	9,850	19,700
0.70	1.74	520	700	870	1,310	1,740	2,610	3,480	5,220	8,710	17,410
0.80	1.56	470	630	780	1,170	1,560	2,350	3,130	4,690	7,820	15,650
0.90	1.42	430	570	710	1,070	1,420	2,140	2,850	4,270	7,120	14,240
1.00	1.31	390	520	650	980	1,310	1,960	2,620	3,930	6,540	13,090
1.10	1.21	360	490	610	910	1,210	1,820	2,430	3,640	6,060	12,130
1.20	1.13	340	450	570	850	1,130	1,700	2,260	3,390	5,660	11,310
1.30	1.06	320	420	530	800	1,060	1,590	2,120	3,180	5,310	10,610
1.40	1.00	300	400	500	750	1,000	1,500	2,000	3,000	5,000	10,000
1.50	0.95	280	380	470	710	950	1,420	1,890	2,840	4,730	9,460
1.60	0.90	270	360	450	670	900	1,350	1,800	2,700	4,490	8,990
1.70	0.86	260	340	430	640	860	1,280	1,710	2,570	4,280	8,560
1.80	0.82	250	330	410	610	820	1,230	1,640	2,450	4,090	8,180
1.90	0.78	230	310	390	590	780	1,170	1,570	2,350	3,920	7,830
2.00	0.75	230	300	380	560	750	1,130	1,500	2,260	3,760	7,520
2.10	0.72	220	290	360	540	720	1,080	1,450	2,170	3,610	7,230
2.20	0.70	210	280	350	520	700	1,040	1,390	2,090	3,480	6,970
2.30	0.67	200	270	340	500	670	1,010	1,340	2,020	3,360	6,720
2.40	0.65	200	260	320	490	650	970	1,300	1,950	3,250	6,500
2.50	0.63	200	250	310	470	630	940	1,260	1,890	3,140	6,290

Table note: For restriction associated with applying these modified values, see **Section 4.5**.

representing future possibilities (e.g., laser diodes), the discrepancy and consequences grow even larger. A 4000-K light source might have an S/P value as low as 0.78 or as high as 5.0.

4.5 Restrictions

The EVE method should only be considered for application when all of these conditions apply:

- The lighting is for an interior application, with photopic viewing conditions
- The visual tasks are covered by IES Illuminance Categories P through Y for the Recommended Illuminance Target (refer to Table A-2 in *ANSI/IES RP-10-20, Recommended Practice: Lighting Common Applications*²⁰)
- The activity is a *visually demanding task*, as defined in **Section 2.5**

- The task is performed under an unobstructed full field of view, with relatively uniform lighting (luminance ratios do not exceed the Default Luminance Ratio Recommendations; refer to Table A-4 in *ANSI/IES RP-10-20, Recommended Practice: Lighting Common Applications*²⁰)
- The lighting is consistent temporally, allowing for complete adaptation
- The task background luminance is 50 cd/m² or greater
- The population is known to have generally good ocular health (typically those under the age of 65)

The EVE method should not be used for the development of energy policy or energy efficiency programs for any lighting applications.

5.0 Implementation Considerations

There are many lighting-based outcomes beyond visual efficiency that are influenced by SPD and illuminance level. Choosing a light source with a higher S/P value and designing to a lower average illuminance level, in accordance with the EVE method, can potentially influence occupants' acceptance of a space by mediating visual comfort, glare, objects' color appearance, spatial brightness perception, and input to non-image-forming visual functions (e.g., circadian synchronization). Some of these aspects may also contribute to task performance, especially glare and color appearance (for chromatic tasks).

5.1 Acceptability and Visual Comfort

It is difficult to isolate the effect of the EVE method on task performance in field studies because tasks vary and are not usually performed at threshold conditions and fixed viewing conditions. However, the acceptability of the resulting illuminated environment has been evaluated in several field studies. A study replacing "835" fluorescent lamps ($S/P = 1.5$) with "850" fluorescent lamps ($S/P = 2.0$) with 20% lower illuminance found no significant difference in occupant acceptance or satisfaction.⁵⁷ A study replacing "730," "735," or "741" fluorescent lamps ($S/P = 1.2$ to 1.6) with reduced-illuminance "850" fluorescent lamps ($S/P = 1.8$ to 1.9) found no significant difference in occupant satisfaction before or after the retrofit.⁵⁸ (In this study, the change in S/P was further confounded with an improvement in color fidelity.) A study replacing existing 3500-K fluorescent lamps (S/P not reported) with 6500-K fluorescent lamps (S/P not reported) at 33% lower illuminance found no statistically significant differences in ratings regarding the overall feeling of the lamps.⁵⁹ In contrast with these three studies, another utilized a factorial design to compare 3500 K ($S/P = 1.4$) and 5000 K ($S/P = 1.9$) at different illuminance levels chosen based on the EVE method. Participants rated the 5000 K conditions too "cool" in appearance and of lower visual comfort.⁶⁰

As with laboratory studies involving the EVE method and visual performance, these field evaluations all share the confound that changes in S/P value occurred simultaneously with changes in CCT—this is not a

necessity, and commercial products exist that can vary melanopic lux without changing CCT. There is a large body of work that has examined preference for various combinations of CCT and illuminance level, in response to the Kruitof curve. A recent review of this body of work found few credible studies and no evidence for changes in preferred CCT with changes in illuminance level.⁶¹

Findings of acceptability and visual comfort may depend on other factors, such as the presence of windows, luminaire luminous intensity distribution, and/or surface spectral reflectance characteristics. Reactions may also depend on the types of tasks being performed, the illuminance level in the space, and other nearby lighting. When considering implementing the EVE method, it is important to consider all aspects of the lighting design.

5.2 Color Rendition

Both illuminance level and SPD influence the color appearance of surfaces in an interior environment. Reducing illuminance makes colors appear duller, which is not preferred in most circumstances.⁶² The illuminance-induced changes in appearance due to the implementation of the EVE method may be smaller than changes to color rendition that occur due to changes in SPD; however, this is difficult to predict and should be considered on a case-by-case basis. Shifts in color appearance will be particularly pronounced if a higher S/P value is achieved by modifying the SPD without increasing the CCT; this practice may also result in duller colors. (For more information, see ANSI/IES LS-5-20⁶³ and ANSI/IES TM-30-18.⁶⁴)

5.3 Spatial Brightness

The effect of SPD on spatial brightness perception is a heavily researched topic, yet there is no model that can successfully capture the effect in all circumstances. A 2013 review found a preponderance of unreliable data, but for the credible studies, a vast majority found the SPD had a significant effect on spatial brightness perception.⁶⁵

Two of the EVE-method field studies previously described investigated spatial brightness perception.^{59,61} In both cases, perceived brightness was equal when higher- S/P sources at lower illuminance were compared

with lower- S/P sources at higher illuminance. However, when sources at equal illuminance but different S/P values were also available for evaluation, there was still no difference in perceived brightness.⁶¹ Another of the field studies⁴⁵ found that a reduction in illuminance significantly increased perceptions that the office appeared gloomy, but this was not the case when S/P was also increased. Again, all of these field studies simultaneously altered S/P and CCT, and in one case also included changes in color rendition.

A few laboratory studies of S/P as a metric for predicting spatial brightness perception have also been published, beginning with the initial evidence in 1990, which looked at SPDs with different S/P values but equal chromaticity.⁶⁶ This finding has subsequently been confirmed, but it was also found that S/P was insufficient to account for variation in spatial brightness perception alone, with a chromatic component not addressed.⁶⁷ Another direct test found that S/P was not a statistically significant predictor of brightness perception.⁶⁸ There is also evidence that the ipRGCs, rather than rods, are a mediator of brightness perception, as with pupillary light reflex.⁶⁹ In summary, the effect of implementing the EVE method on spatial brightness is indeterminate because the physiological mechanisms underlying the two phenomena are different. More information on spatial brightness is provided in ANSI/IES LS-8-20.¹

5.4 Glare

As with visual performance, both SPD and luminous intensity influence perception of glare. Increases in CCT (and S/P) have been linked to increased glare,⁷⁰⁻⁷³ although the SPD/illuminance tradeoff of the EVE method has not been tested. Importantly, pupil size is correlated with discomfort glare, with smaller pupil size occurring with increased discomfort from glare.^{44,74} Increasing S/P without an accompanying decrease in illuminance has the potential to increase glare; however, glare depends on many factors beyond SPD and luminous intensity. Glare can also influence visual performance, so care should be taken to understand all of the effects of changing SPD and illuminance when implementing the EVE method.¹

More information on glare is provided in ANSI/IES LS-8-20.¹

5.5 Other Physiological Responses

The ipRGC is linked to many non-image forming responses to light, which are important regulators of physiology and behavior.⁴⁷ Changes in SPD occurring with implementation of the EVE method can influence outcomes such as circadian synchronization and acute alerting effects. Most non-image forming responses to light have been shown to have a complex phototransduction pathway that extends beyond a single photoreceptor, so changes to illuminance and S/P values in accordance with the EVE method will have variable effects, potentially both positive and negative. These considerations will require accurate phototransduction models in order to understand the tradeoffs that occur as SPD varies. Research in this field is very active, and the state of knowledge is constantly changing.

Advances in understanding of circadian synchronization and light's effect on wellbeing have suggested the need for higher melanopic flux during the day in many circumstances. The amount of short-wavelength energy required to aid circadian entrainment is often too great to achieve through manipulation of the SPD alone, instead requiring concurrent increases in overall radiant power—which also increases delivered illuminance. That is, to address the photobiological needs of building occupants often requires substantially higher illuminance levels than when designing for visual performance alone (using standard IES illuminance recommendations). Utilizing an SPD/illuminance tradeoff, as prescribed in the EVE method, does not lead to higher melanopic watts. More work is needed to understand how the many human response influence by SPD relate to one another.

Additional information may be found in *IES TM-18-18, Light and Human Health: An Overview of the Impact of Optical Radiation on Visual, Circadian, Neuroendocrine, and Neurobehavioral Responses*.⁷⁵

5.6 Daylighting

Where windows are present, daylight has an effect on the overall spectral composition of lighting within interior environments. The actual SPD of the daylight contribution depends on a number of variables, including time of day, latitude, building orientation,

and fenestration spectral transmittance. To the extent that daylight is available, it generally provides higher S/P values and therefore might contribute to visual efficiency.

6.0 Future Considerations

The EVE method was first formalized in 2013⁸ and has been successfully implemented by some practitioners. Like most methods in applied lighting, it is subject to continued improvement. In particular, the $P(S/P)^{0.78}$ function, leading to the EVE method, is based on an empirical fit and inaccurate physiology corresponding to the multi-input phototransduction pathway that controls the pupillary light reflex. A complete model of the pupillary light reflex could improve the external validity of the EVE method, ensuring that its use with a wide variety of spectral power distributions provides the intended result.

It is well established that eyes for a large segment of the population are known to retain some residual refractive error. In addition, it is also well known that there will generally be an acuity benefit for such eyes resulting from reductions of pupil size, which allow a sharper retinal image. Reducing pupil size can be achieved by raising light level, but a central theme of this document is that similar reductions can also be achieved by spectral considerations without raising light level. Based on these features, and for the purposes of this document, environmental lighting conditions that yield, on average, equal visual acuity are assumed to have equal visual efficiency. Furthermore, it is proposed on the basis of studies described in IES TM-24-13,⁸ Annex B, that the principal driver of acuity for the task conditions considered herein is the steady-state or tonic pupil size, which is primarily controlled by the overall lighting conditions. It is also acknowledged that pupil size can be modulated by other physical and psychological parameters, but it is assumed here that the light-driven steady state size, via its connection to visual acuity, adds a useful tool to this area of lighting practice.

To date, the vast majority of experiments involving the EVE method and broader considerations of the illuminance-level/SPD tradeoff have been conducted using narrowband fluorescent lamps with different S/P values *and* different CCTs. Thus, they have simultaneously varied multiple photoreceptor inputs. Given the increasing use of LEDs for architectural lighting, explicit verification of the EVE method to LEDs is warranted. LED technology also facilitates creation of visual metamers with varied input to the pupillary light reflex, which can be used to build a more robust model of the illuminance-level/SPD tradeoff.

There is growing evidence of signaling from the different combinations of photoreceptors that project to many nuclei in the brain, thus influencing, for example, circadian rhythms, alertness, attention, and decision making, some of which could influence visual performance beyond regulating pupil size. A more complete understanding of these mechanisms and models built from neurology in addition to empirical fits could improve this and other methods used in illuminating engineering.

Additional work to understand the range of conditions that are acceptable for using the EVE method for maintaining visual efficiency of visually demanding tasks in real environments is warranted. New work could result in more-relaxed or more-tightened restrictions for when the EVE method can be used.

While the research on spectrally driven visual acuity has been limited to two-dimensional tasks, improved depth perception is also a consequence of reduced pupil size. This suggests that the spectral concept could be applied to three-dimensional tasks as well, such as those in sports and industrial activities. However, this has not been verified.

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**An Optional Method for Adjusting the Recommended Illuminance for Visually Demanding Tasks
within IES Illuminance Categories P through Y Based on Light Source Spectrum**

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
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- ☐ Add new text as follows
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Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

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6. Reason and substantiation:

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Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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